

THE LANGUAGE OF BUSINESS ANALYTICS

1. Foundational Concepts OF Analytics

1. BUSINESS ANALYTICS

Refers to the statistical methods and computing technologies for processing, mining and visualizing data to uncover patterns, relationships and insights that enable better business decision making.

Business Analytics makes extensive use of analytical modeling and numerical analysis, including explanatory and predictive modeling to drive decision making. It may be used as input for human decisions.

Business Analytics can answer questions like why is this happening, what if these trends continues, what will happen next (predict) and what is the best outcome that can happen (optimize)

REAL - WORLD BUSINESS ANALYTICS EXAMPLES

A Supermarket wants to reduce food waste.

- A Supermarket will then collect data about, Daily Sales, Stock levels, Customer buying patterns, product prices, expiry dates & seasonal demand.
- Using business Analytics, the supermarket can discover that certain products sell slowly on particular days. The Supermarket can reduce waste, lower costs and improve profits.

REFERENCES

1. <https://www.ibm.com>
2. <https://www.knowledgehut.com>
3. <https://www.researchmethod.com>

2. DATA ANALYTICS

Data Analytics is the process of collecting, cleaning, organising, examining and interpreting data to discover useful information, patterns, trends and relationships.

It helps Individuals and organisations understand what has happened, why it happened, what may happen in the future, what actions could be taken based on the evidence in the data.

REAL-WORLD EXAMPLE : Netflix

Netflix collects large amounts of data about its users interact with the platform.

It can analyse:

- which movies or TV shows users watch,

- How long users watch a particular programme.

- Which programmes users stop watching before finishing.

- What genres users frequently select.

- Whether users rewatch particular programmes.

Netflix can analyse this data to identify viewing patterns and customer

3. BUSINESS INTELLIGENCE

It is a set of technological processes for collecting, managing and analyzing Organizational data to yield insights that inform business strategies and operations.

Business Intelligence turns business data into information that managers can use to make decisions.

REAL-WORLD EXAMPLE

Healthcare Industry.

The healthcare industry generates a massive amount of data daily, from patient records to treatment protocols. BI helps by:

- Improving patient care, hospitals use BI to predict which patients are at risk of readmission.
- Resource Management, Data analytics can forecast peak times and ensure the availability of necessary resources.
- Cost Management, Controlling expenses by identifying inefficiencies in workflow processes and patient management systems.

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1. <https://www.netsuite.com>
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4. DATA-DRIVEN DECISION-MAKING

Is an approach that emphasizes using data and analysis instead of intuition to inform business decisions.

It involves leveraging data sources such as customer feedback, market trends and financial data to guide the decision making process.

By collecting, analyzing and interpreting data, Organizations can make better

decisions that more closely align with business goals and objectives.

REAL-WORLD EXAMPLE

A Supermarket

A Supermarket notices that sales of milk have decreased.

Management could assume that customers no longer want the product.

Instead, they examine the data and it shows that:

- Milk sales have decreased by 20%.
- Several stores frequently run out of milk.
- Customers are still purchasing milk when it is available.
- Competitors' prices have remained similar.

The business can therefore investigate stock availability rather than immediately changing the product or lowering its price.

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1. <https://www.ibm.com>
2. <https://www.saber.app>
3. <https://www.numberanalytics.com>

5. BIG DATA

Big data refers to massive, complex datasets that traditional data management systems cannot handle. When properly collected, managed and analyzed, big data can help organizations discover new insights and make better business decisions.

EXAMPLE OF BIG DATA

IN FINANCE.

Fraud detection, financial houses implement Big Data to deal with fraud. Algorithms do analysis of patterns and flag consistently any fraudulent activity, which in turn reduces fraud cases by a large percentage and saves the institution millions of rands every year.

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1. <https://www.geeksforgeeks.org>

2.

6. DESCRIPTIVE ANALYTICS

Descriptive analytics is a data analysis approach focused on summarizing and interpreting historical data to understand what happened and what is currently happening. It uses statistical summaries and data visualization to identify patterns, trends and relationships, forming the foundation for deeper analytics like diagnostic, predictive methods.

EXAMPLE OF DESCRIPTIVE ANALYTICS

A hospital analyzing patient records. They might calculate average wait times in the emergency room, categorize the most frequent diagnoses, or track year over year trends in admissions. By summarizing data and using visuals like bar charts, line graphs, they can identify patterns and understand what's happening within the hospital. This allows them to focus on areas for improvement like reducing wait times.

REFERENCES

Service or unlikely to renew, so the company can personalise marketing offers or customer education efforts to encourage them to stay.

REFERENCES

1. <https://www.sap.com>
2. <https://researchmethod.net>
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4. <https://www.sas.com>

9. PRESCRIPTIVE ANALYTICS

It is a type of data analytics that seeks to understand what is needed to achieve an aim. Prescriptive analytics uses data analytics with technology to enhance business decisions, by recommending strategies based on performance metrics and scenarios.

It works with predictive analytics to suggest optimal actions.

EXAMPLE OF PRESCRIPTIVE ANALYTICS

Uber's Ride Optimization, Uber applies prescriptive analytics to improve ride efficiency. It uses real-time data to match riders to drivers, considering factors like traffic and past ride patterns to suggest optimal routes. Uber also predicts demand during events like concerts, guiding drivers to high demand areas which reduces wait times for riders and increases earnings for drivers.

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3. <https://researchmethod.net>
4. <https://investopedia.com>

10. DATA LITERACY

Data literacy is the ability to explore, understand and communicate with data in a meaningful way.

11. STRUCTURED DATA

Structured data is data that is organised in a clear, predefined format, usually in rows and columns, so that computers can easily store, search, analyse, and process it.

EXAMPLE OF STRUCTURED DATA

Financial transactions such as credit card transactions, bank deposits, and wire transactions/transfers are all examples of structured data. Each transaction comes with specific information in the form of a serial number, a transaction date, the amount, and the parties involved. This information is structured and stored in databases enabling banks and financial institutions to track and analyze financial activities.

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1. <https://www.datamation.com>
2. <https://www.simplicable.com>
3. <https://estuary.dev>
4. <https://marketingscoop.com>

12. UNSTRUCTURED DATA

Unstructured data is data that does not have a predefined or organised format, such as rows and columns in a traditional database. It is information that doesn't fit neatly into a table or a fixed structure.

REAL WORLD EXAMPLE

- Emails are among the most popular unstructured data examples we use everyday for business or personal purposes.
- Websites like YouTube, photo sharing sites, Instagram are also examples of unstructured information. All web content is considered as a source of raw and unstructured information.

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2. <https://www.ibm.com>
3. <https://www.datamation.com>
4. <https://www.expertbeacon.com>
5. <https://>

13. DATA WAREHOUSE

A data warehouse is a central system where a company stores large amounts of organised data collected from different sources. Used mainly for analysis & reporting, and stores historical data from different sources.

EXAMPLE OF DATA WAREHOUSE

A Supermarket such as Shoprite collects sales data from many stores and stores it in a data warehouse. The company can then analyse the combined data to see, which products sell the most, which stores have the highest sales and customer buying patterns.

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2. <https://www.ibm.com>
3. <https://azure.microsoft.com>
4. <https://en.wikipedia.org>

14. DATA MINING

Data mining is the process of analyzing large datasets to uncover hidden patterns, relationships, and valuable insights. It uses machine learning and statistical analysis to uncover patterns and other valuable information from large data sets.

REAL-WORLD EXAMPLE

A supermarket can use data mining to analyse thousands of customer transactions and discover that customers who frequently buy bread also buy milk.

The supermarket can use this pattern to create promotions, product recommendations or place products closer together.

~~REFERENCE~~

REFERENCE

1. <https://www.ibm.com>
3. <https://www.marketingcoop.com>

15. DATA VISUALISATION

Data visualization is the graphical representation of information and data using visual elements like charts, graphs and maps.

REAL WORLD EXAMPLE

CDC COVID-19 Hospitalization Trackers: Public health groups use color coded line graphs and regional maps to display disease spread, peak infection times and hospital capacity.

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1. www.ibm.com
2. www.simplifiable.com

16. DASHBOARD

A dashboard is a tool for monitoring and displaying real time data.

REAL WORLD EXAMPLE

Manufacturing Plant Dashboard: Visualizes real time shop floor metrics like equipment downtime, production throughput, and actual versus standard output efficiency.

REFERENCE

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2. <https://www.tableau.com>
3. <https://www.activityinfo.org>

17. METRICS

Metrics are quantifiable standards and sets of numbers used to measure, track and evaluate the performance, status or progress of a specific process, activity or goal.

REAL WORLD EXAMPLE

YouTube Metrics:

Growth Metrics,

- Monthly/Daily Active Users: How many people watch YouTube videos regularly.
- New Users: The number of new accounts created each month.

Engagement Metrics,

- Video Views per users: The average number of videos each user watches.
- Average Viewing Time: How long users stay watching videos.

18. KEY PERFORMANCE INDICATOR (KPI)

KPI are quantifiable measures used to evaluate progress toward specific business objectives. They can be financial or non financial, depending on what the organisation is trying to achieve.

REAL WORLD EXAMPLE

Customer Support: An online store wants to make shoppers happier by answering support questions faster.

The Goal is to improve customer satisfaction by reducing wait times.

The KPI, Average Resolution Time, the average time it takes to fix a customer problem from start to finish.

The Target is to fix every issue in under 4 hours.

The Result is if the team average drops from 6 hours to 3.5 hours, the KPI shows the goal is met.

REFERENCE

19. BENCHMARK

Refers to a standard or point of reference against which other things can be measured or compared. It is a tool to evaluate performance, set goals and make informed decisions.

REAL WORLD EXAMPLE

Customer Service Call Centres

A call center tracks its average waiting time and first contact resolution rate, then compares these metrics against its industry competitors to find gaps and train staff.

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3. <https://www.euruni.edu>

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20. SEGMENTATION

Is the strategic practice of dividing a broad target market or customer base into smaller, distinct groups based on shared characteristics, needs or behaviours.

REAL WORLD EXAMPLE

Netflix, Behavioral Taste Clusters.

Rather than focusing on where a viewer lives or how old they are, Netflix groups its audience into ultra specific "Taste Clusters" based entirely on viewing habits. Netflix isolates Weekend Bingers from Casual browsers. Weekend bingers are immediately fed deep multi season shows, whereas casual users are recommended shorter films or limited series.

21. CONVERSION RATE

Conversion rate is the percentage of people who take a desired action out of the total number of visitors or interactions.

REAL WORLD EXAMPLE

- SaaS (Software as a Service), A project management tool gets 10,000 visitors to its landing page and 300 people sign up for a free 14 day trial. The Conversion rate is 3%.
- A cooking blog has 20000 weekly readers and 1000 people subscribe to its weekly recipe email newsletter. The Conversion rate is 5%.

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2. <https://www.geckoboard.com>
3. <https://www.agencyanalytics.com>
4. <https://www.hireawriter.us>
5. <https://www.visionlabs.com>

22. CUSTOMER ACQUISITION COST (CAC)

The total expense a Company incurs to acquire a new Customer, calculated by dividing all Sales and marketing costs by the number of new Customers gained over a specific period.

EXAMPLE

Imagine Netflix Spends R1000 000 on advertising and marketing in one month to attract new Customers. During that month, the marketing campaign brings in 20 000 new Customers.

The CAC would be : $CAC = \frac{\text{Total Marketing and Sales Costs}}{\text{Customers}} \div \text{no of new}$

$$CAC = R1000000 \div 20000 = R50$$

This means Netflix Spent an average of R50 to acquire each new Customer through its marketing activities.

23. CUSTOMER LIFETIME VALUE

Is the total amount of money a business expects to earn from a customer remains a customer during the entire period that the customer remains a customer.

EXAMPLE

Businesses use CLV to understand how valuable customers are and to decide how much they can reasonably spend to acquire and retain customers.

24. CHURN RATE

Is a percentage of customers who stop using a company's product or service during a specific period of time.

EXAMPLE

Suppose Netflix has 10 000 customers at the beginning of a month. During that month, 500 customers cancel their subscriptions.

Churn Rate = $500 \div 10\,000 \times 100 = 5\%$, This means that 5% of the customers Netflix had at the beginning stopped using the service during that month.

25. CUSTOMER RETENTION RATE

Is a percentage of customers that a business keeps over a specific period of time.

EXAMPLE

Suppose Netflix has 10 000 customers at the beginning of the month, 10 500 at the end, and new customers gained is 1 500.

Retention Rate = $[(10\,500 - 1\,500) \div 10\,000] \times 100 = 90\%$

Therefore, Netflix customer retention rate is 90% for the month.

26. NET PROMOTER SCORE

Is a metric businesses use to measure how likely customers are to recommend a Company's product or service to other people.

EXAMPLE

Suppose Netflix surveys 1000 customers, 600 customers give a score of 9-10, 60% promoters. 250 customers give a score of 7-8, 25% passives. 150 customers give a score of 0-6, 15% Detractors.

$$NPS = 60\% - 15\% = 45$$

27. AVERAGE ORDER VALUE

Is the average amount of money a customer spends on each order or transaction.

EXAMPLE

Suppose Takealot makes R100 000 in sales from 2000 orders in one month.

$$AOV = R100\,000 \div 2000 = R50, \text{ therefore the average AOV is R50,}$$

This means that an average customers spend R50 per order.

28. SALES FUNNEL

Is the step by step process a potential customer goes through from first becoming aware of a business to eventually making a purchase.

EXAMPLE

Someone sees a Takealot advertisement for a laptop on social media, they click on it and look at the laptop, they compare the prices, specifications and reviews with other laptops, they add it to the cart and pay. Takealot sends them personalised recommendations or promotions that encourage them to shop again.

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3. <https://www.techtarget.com>

2. <https://www.salesforce.com>

29. MARKETING CAMPAIGN

It is a planned set of marketing activities designed to achieve a specific business goal, such as increasing sales, attracting new customers, launching a product, or improving brand awareness.

EXAMPLE

Showmax could run a campaign to promote a new series. It might create trailers and short videos for social media, send emails to subscribers recommending the series, use advertisement on YouTube and other platforms. The goal could be to increase viewership and engagement with the new series.

30. POST-CAMPAIGN ANALYSIS

Is the process of reviewing a marketing campaign after it has finished to determine whether it achieved its objectives and to understand what worked, what did not work, and what should be improved in future campaigns.

EXAMPLE

Suppose Netflix promotes a new series through social media and email, Netflix sets a goal of getting 1 million additional viewers. After the campaign, Netflix analyses the results and finds that, 10 million people saw the adverts, 2 million visited the series page, 1.2 million watched the series, Netflix can use this information to determine whether the campaign achieved its objectives and which marketing channels generated the strongest results.

31. INCREMENTAL REVENUE

Is the additional money a business earns as a direct result of a specific action, campaign, product launch, promotion or change compared with what it would have earned without that action.

EXAMPLE

Netflix launches a special promotional campaign to attract new subscribers.
Revenue before the campaign: R10 million,

Revenue after the campaign, R12 million per month.

Incremental Revenue = R12 million - R10 million = R2 million. Therefore the campaign generated R2 million in additional revenue.

32. RETURN ON INVESTMENT

Is a financial measure used to determine ~~used~~ how much profit or return a business receives compared with the amount of money it invested.

EXAMPLE

Suppose a company spends R100 000 on a digital marketing campaign. The campaign generates R150 000 in revenue.

$$\text{Net Return} = R150\,000 - R100\,000 = R50\,000$$

$\text{ROI} = (R50\,000 \div R100\,000) \times 100 = 50\%$, This means the company generated a 50% return on its investment.

33. RETURN ON AD SPEND (ROAS)

Is a marketing metric that measures how much revenue a business generates for every rand it spends on advertising.

EXAMPLE

A clothing store spends R20 000 on an Instagram and Facebook advertising campaign. The company generates R80 000 in sales.

$\text{ROAS} = R80\,000 \div R20\,000 = 4$. This means the business generated R4 in revenue for every R1 spent on advertising.

34. CLICK THROUGH RATE

Is a marketing metric that measures the percentage of people who click on an advertisement, email link, search result or other digital link after seeing it.

EXAMPLE

An online shoe store runs a Google advertisement for a new pair of running shoes.

The advertisement is shown 10 000 times, 500 people click on it.

$CTR = (500 \div 10000) \times 100 = 5\%$, Therefore, the advertisement has a 5% Click-through rate. This means 5% of people who saw the advert clicked on it.

35. BOUNCE RATE

Is a digital marketing and website metric that measures the percentage of visitors who leave a website after viewing only one page without taking any further action.

EXAMPLE

An online clothing store receives 1000 website visits in one day. Out of those visitors, 400 leave after viewing only the page they initially visited.

$$\text{Bounce Rate} = (400 \div 1000) \times 100 = 40\%$$

Therefore the website has a 40% bounce rate for that period.

36. A/B TESTING

Is a method of comparing two different versions of something, such as an advert, webpage, email, headline or button to determine which version performs better.

EXAMPLE

An online clothing store wants to find out which "Buy Now" button works better. Version A: Buy Now and Version B: Shop Now, different groups are shown these versions. The company then collects enough results, Version A receives 100 purchases and Version B receives 140 purchases.

The business can then use the results to understand which version produced more purchases.

37. LEAD GENERATION

Is the process of attracting and identifying people who may be interested in a company's product or service and turning them into potential customers (leads).

EXAMPLE

If Discovery Health runs a Facebook advertisement offering information about medical aid plans. A person sees the advert and clicks "Get quote". They enter their name, email address and phone number.

Discovery now has a lead because that person has shown interest and can potentially become a customer.

38. MARKETING ATTRIBUTION

Is the process of determining which marketing activities or channels contributed to a customer becoming a lead or making a purchase. It helps a business understand where its customers came from and which marketing efforts influenced their decision to buy.

EXAMPLE

A customer buys a pair of shoes from an online store. Before purchasing, the customer, sees the company's Instagram advertisement, searches for the company on Google, visits the website, receives an email promotion and clicks the email and purchases the shoes. Marketing attribution helps the business analyse these different interactions and determine how each marketing channel contributed to the sale.

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